

PERFORMANCE TESTING

Project: Fitbuddy AI Fitness Plan Generator – Gemini Models

Date: 30 September 2026 | Maximum Marks: 5

Team Details

Role	Name
Team Leader	Abisreeya S
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Purpose: Performance testing evaluates how Fitbuddy behaves under expected and peak load, including the time taken to generate a personalized fitness plan and the stability of the service.

Step 1: Testing Overview

Testing Tool Used	Apache JMeter (proposed tool; replace if another tool is used)
Type of Testing	Load testing, stress testing, and spike testing
Target Module / API	Fitness-plan generation request and Gemini model integration
Test Environment	Development / test environment; record the actual hosting setup before testing
Test Date	30 September 2026 (planned/documentation date; update to the actual test date)

Step 2: Test Scenarios

S.No	Test Scenario / Description	Virtual Users	Duration (sec)	Expected Outcome
1	Normal load: users submit fitness goals and request a plan	10	60	Requests complete reliably; average response time target is under 2 seconds where feasible
2	Increasing load: gradually increase concurrent plan-generation requests	25	120	Response time and resource use remain within defined limits
3	Stress test: increase users beyond expected peak load	50	120	System degrades gracefully and reports errors without crashing
4	Spike test: sudden burst of plan-generation requests	10 → 50	60	Service recovers after the spike; no sustained failure

Note: Virtual-user counts and durations above are suggested sample test settings, not measured results. Adjust them to match expected usage and the available test environment.

Step 3: Performance Test Results

Actual measurements have not been supplied. Fill in the Actual Value and Status columns after running the test; do not mark a metric Pass without evidence.

S.No	Metric	Target Value	Actual Value	Status / Remarks
1	Average response time	< 2 seconds	To be measured	Pending test
2	Maximum response time	< 5 seconds	To be measured	Pending test
3	Throughput	Record requests/second; set project target	To be measured	Pending test
4	Error rate	< 1%	To be measured	Pending test
5	CPU utilization	< 80%	To be measured	Pending test
6	Memory utilization	< 80%	To be measured	Pending test

Step 4: Observations & Analysis

To be completed after executing the performance tests. Review response-time trends, throughput, error rate, CPU and memory usage, Gemini API latency, rate-limit responses, and recovery after peak load. Record bottlenecks and any changes made.

Observed bottlenecks: _____

Gemini/API latency or rate-limit behavior: _____

Actions taken / improvements: _____

Overall result: Pending actual test execution

Step 5: Screenshots / Evidence

Attach JMeter summary reports, response-time charts, throughput graphs, error logs, and CPU/memory monitoring screenshots after testing. No screenshots or measured evidence were provided with this document.

Evidence file / screenshot 1: _____

Evidence file / screenshot 2: _____